

Chapter Six: Waves in the Atmosphere

- ❑ The Perturbation Method

- ❑ Properties of Waves

- ❑ Types of Waves

 - Acoustic Waves

 - Gravity Waves

 - Rossby Waves

Introduction

- Wave motions are oscillations in field variables (such as velocity and pressure) that propagate in space and time.
- Waves in fluids result from the *action of restoring forces* on fluid parcels that have been displaced from their equilibrium positions.
- The *restoring forces* may be due to *compressibility, gravity, rotation, or electromagnetic effects*.
- Wave motion is responsible for the transport of *large amounts of energy and momentum in several atmospheric regimes of circulation*.

The Perturbation Method

- The perturbation method is a simple technique that is useful for qualitative analysis of atmospheric waves.
- In the perturbation method, all field variables are divided into *two* parts:
 - *basic state*: assumed to be independent of time and longitude
 - *perturbation*: the local deviation of the field from the basic state.
- Example: $u(x, t) = \bar{u} + u' (x, t)$ (complete zonal velocity field)
 - $\bar{u}$ is averaged zonal velocity
 - u' is the deviation from the average

The Perturbation Method

- For example, $u \frac{\partial u}{\partial x}$ can be written as:

$$u \frac{\partial u}{\partial x} = (\bar{u} + u') \frac{\partial}{\partial x} (\bar{u} + u')$$

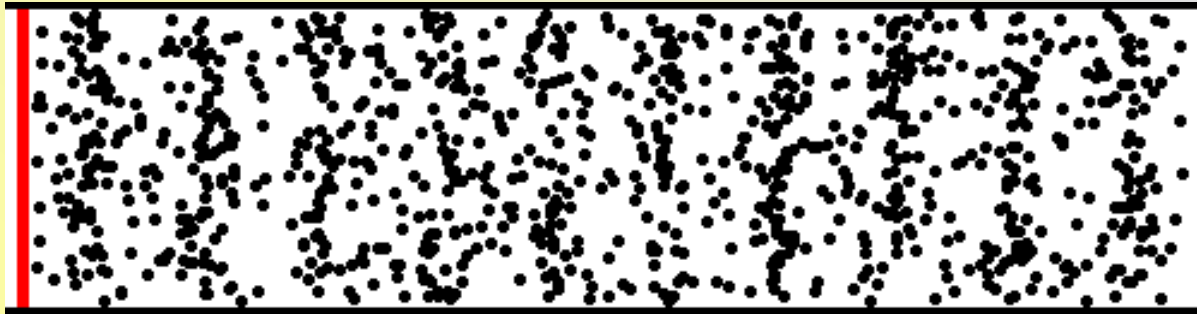
- The basic assumptions of perturbation theory are:
 - the basic state variables must themselves satisfy the governing equations when the perturbations are set to zero
 - the perturbation fields must be small enough so that all terms in the governing equations that involve products of the perturbations can be neglected (i.e. $\frac{u'}{\bar{u}} \ll 1$)

Properties of Waves

Longitudinal Wave

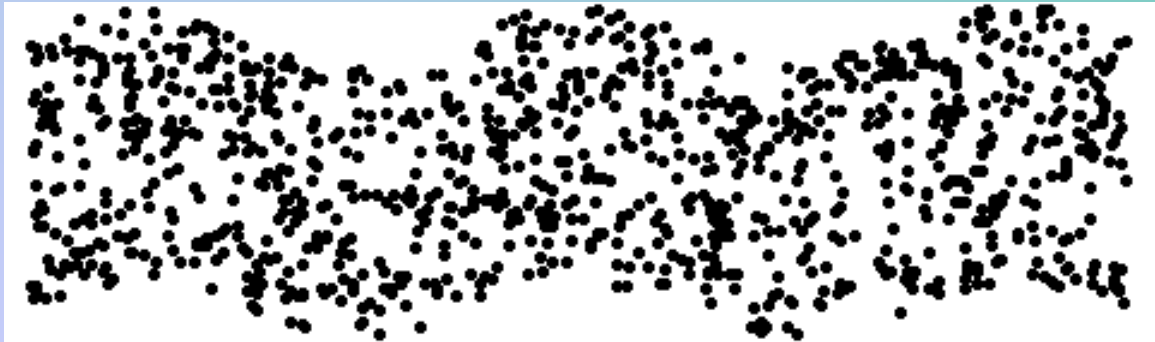
wave particles vibrate back and forth along the path that the wave travels.

Compressional Wave



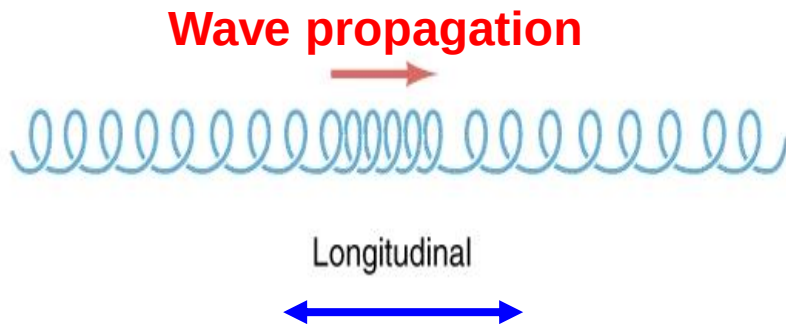
Transverse waves

wave particles vibrate
in an up-and-down motion.



Transverse and Longitudinal Waves

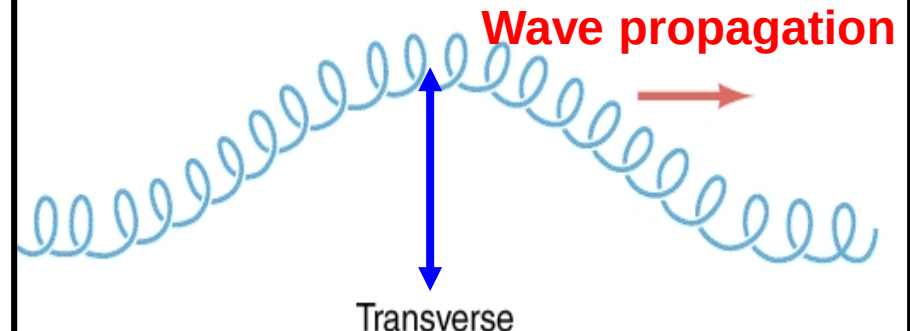
- Longitudinal wave
- Compression
- The oscillation is perpendicular the phase lines
- Example: Sound wave



10/24/2024 **Particle oscillation**

By Israel G.

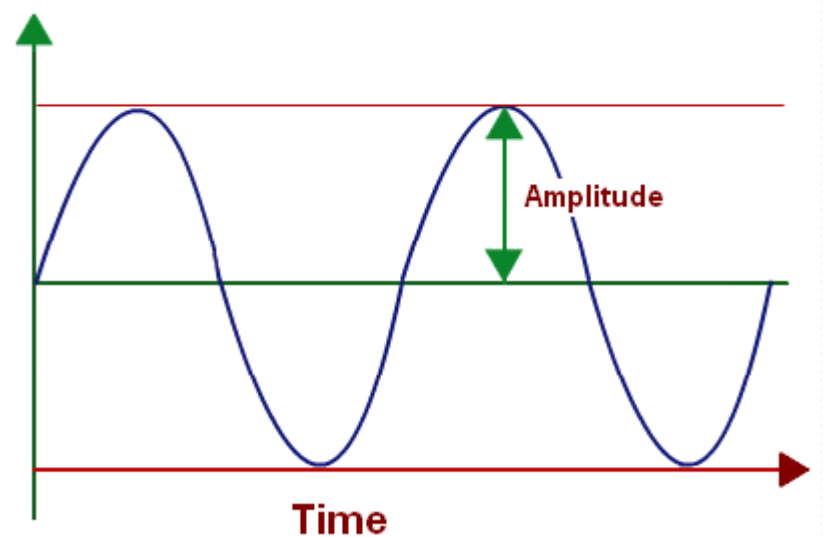
- Transverse wave
- No compression
- The oscillation is along the phase lines
- Example: Gravity wave



Particle oscillation

Wave Characteristics

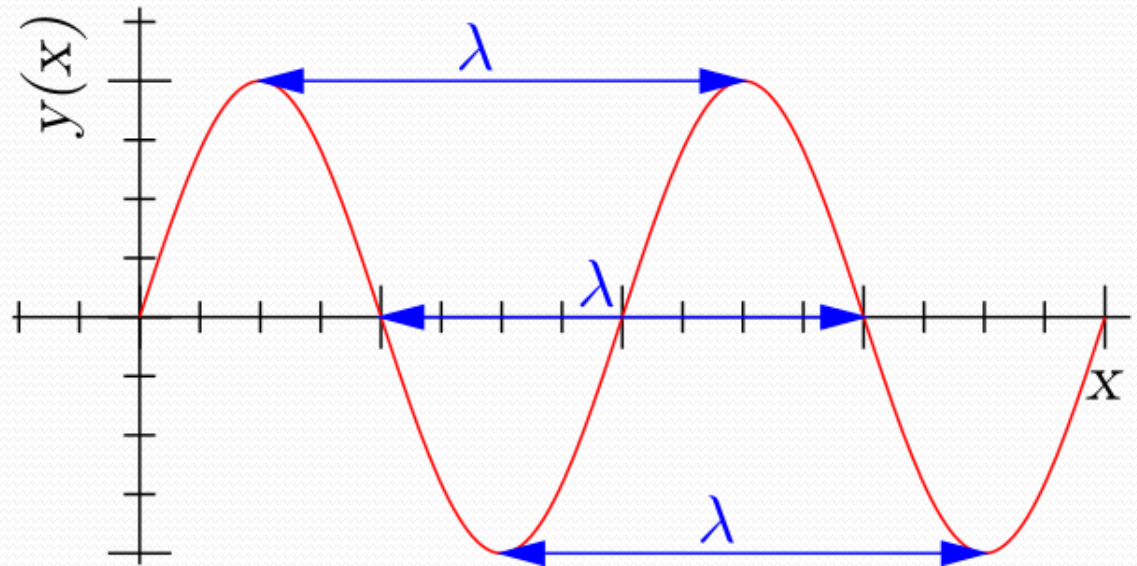
- **Amplitude:** The greatest distance from equilibrium.
- The bigger the amplitude of the wave, the more energy the wave has.



Wave Characteristics

- **Wavelength:** The shortest distance between points where the wave pattern repeats itself.

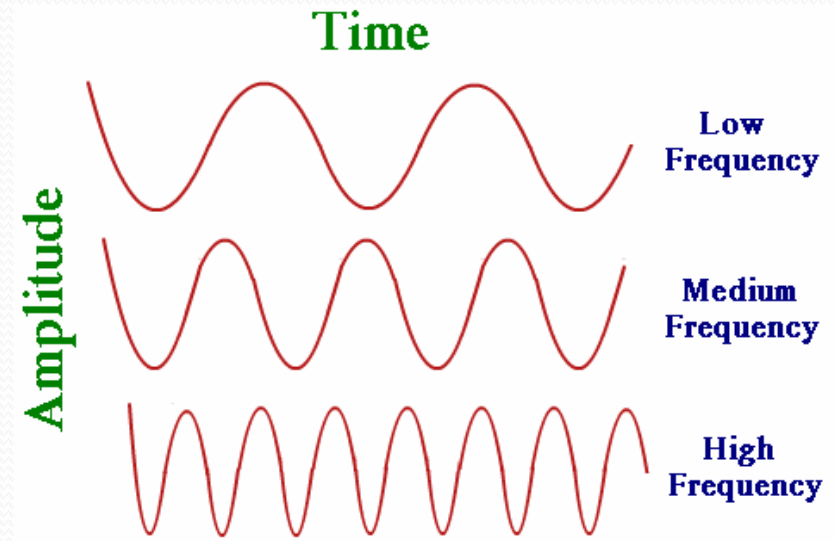
Wavelength is measured in meters, the symbol is λ



Wave Characteristics

- **Frequency:** Is the number of complete oscillations a point on that wave makes each second.

Frequency is measured in Hertz (Hz), its units are $\frac{1}{s}$



Wave Characteristics

- **Period:** The time it takes a wave to go through one cycle, or the time it takes a point to go through one phase of the wave.
- The period of a wave is measured in seconds, and it's symbol is “T”.
- $T = \frac{1}{f}$

Wave Characteristics

- **Wave Velocity** - is the speed with which a wave crest passes by a particular point in space
- It is measured in meters/second.
- $\text{Wave Velocity} = \text{Frequency} \times \text{Wavelength}$

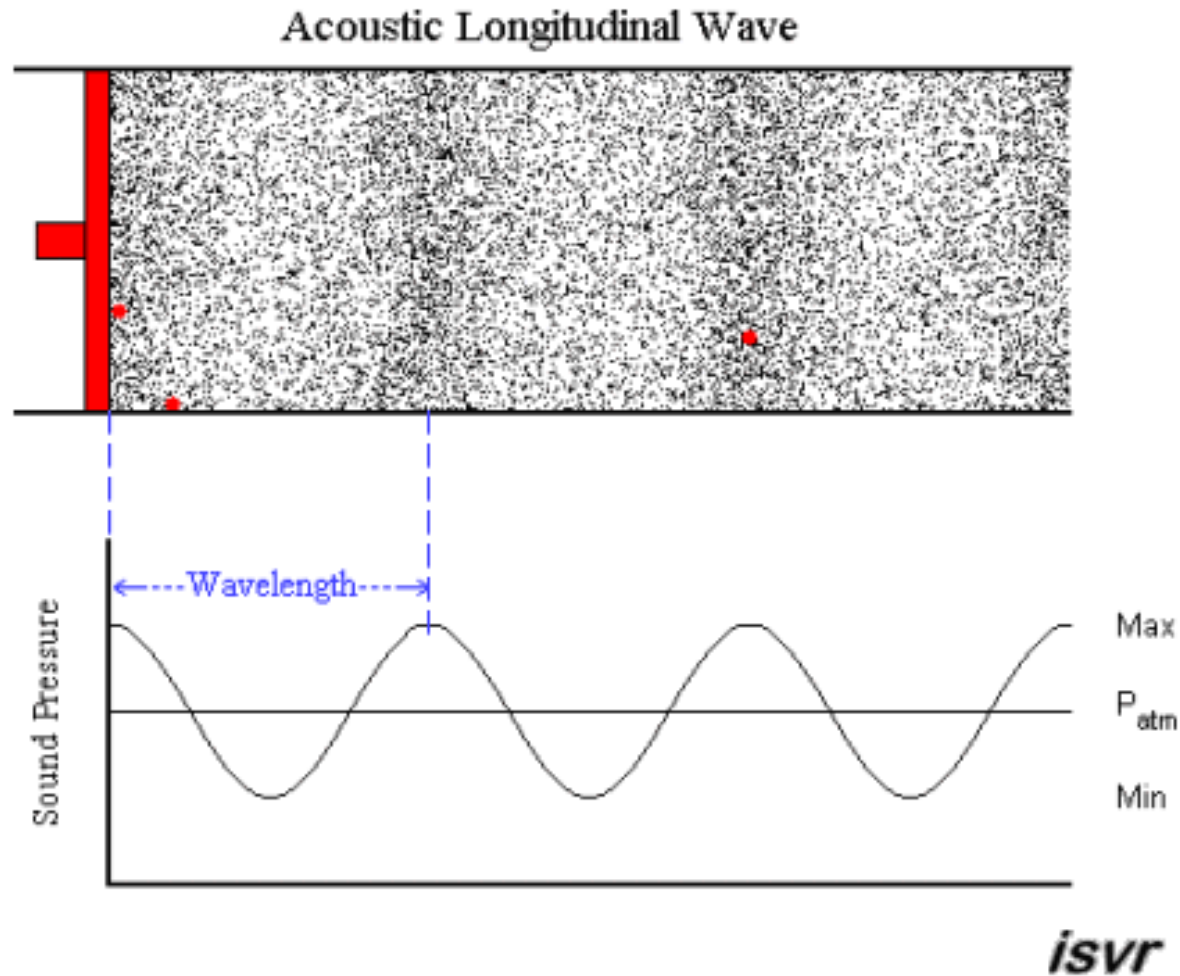


Types of Waves

Acoustic or Sound Waves

- Sound waves, or acoustic waves, are *longitudinal waves* (the particle oscillations are parallel to the direction of propagation)
- Sound is propagated by the alternating adiabatic compression and expansion of the medium.
- The continuity equation has to play an important role

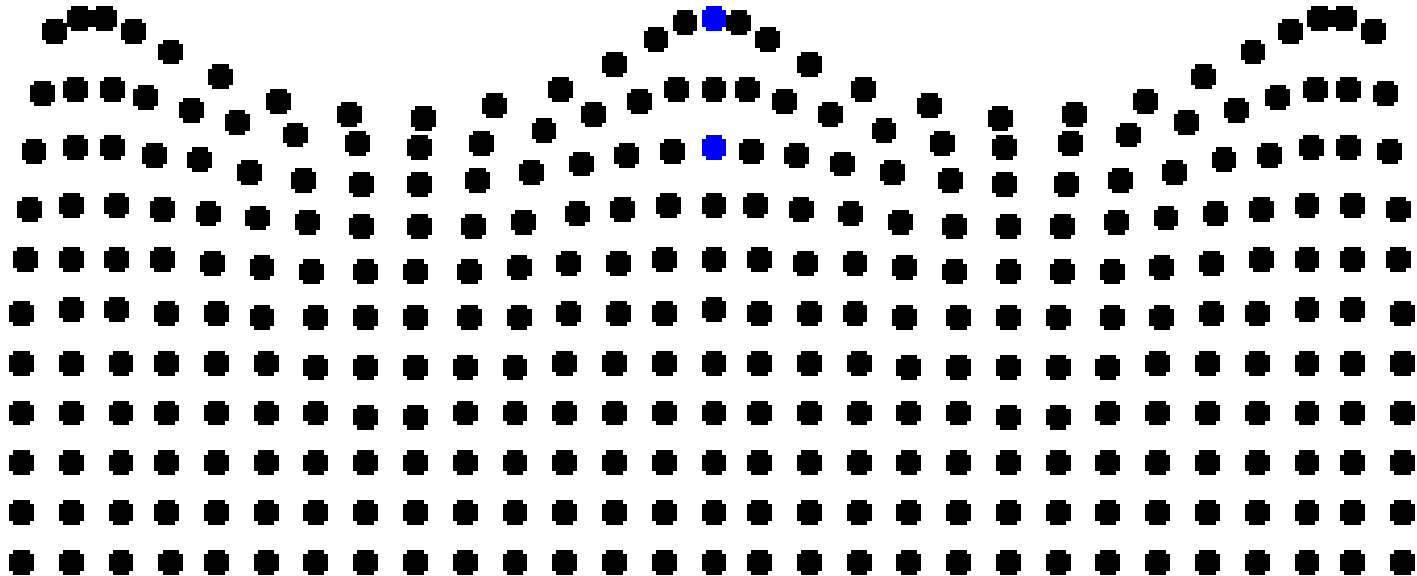
Sound Waves



Gravity Waves

- Gravity waves are buoyancy waves – the restoring force comes from Archimedes's principle.
- The restoring force is in the vertical so that it is transverse to the direction of propagation.
- They can propagate vertically and horizontally, transporting momentum from their source to their sink

Gravity Waves



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Rossby Waves

- Rossby waves are the *time-dependent motion of the large-scale atmosphere.*
- The wave type that is of most *importance for large-scale meteorological processes* is the Rossby wave, or planetary wave.
- The rotation of the earth is important
- Typical sources: large topography like continents; land-sea thermal contrast;

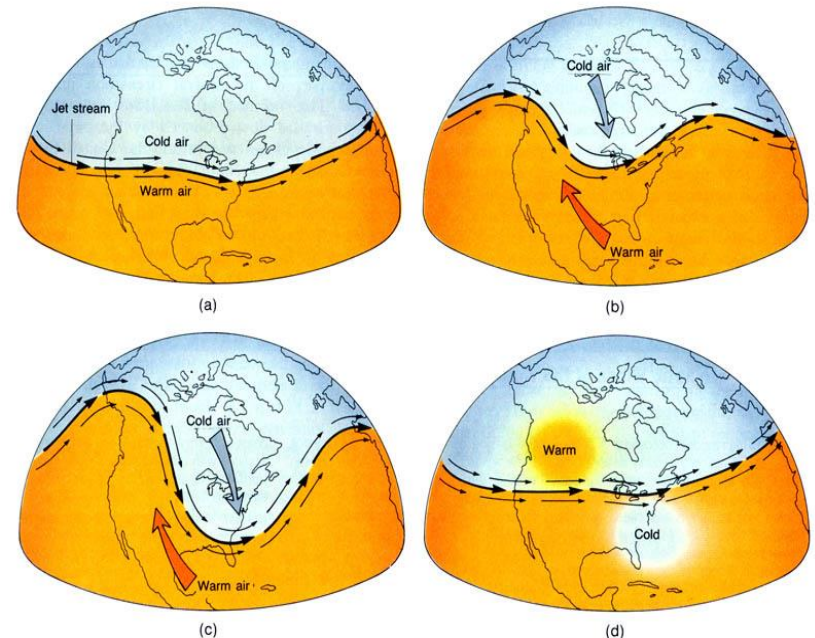


Figure 8•9 Cyclic changes that occur in the upper-level airflow of the westerlies. The flow, which has the jet stream as its axis, starts out nearly straight and then develops meanders that are eventually cut off. (After J. Namias, NOAA)

1. The wave type that is of most importance for large-scale meteorological processes
 - A. Gravity wave
 - B. Sound wave
 - C. Rossby wave
 - D. None
2. is a simple technique that is useful for qualitative Analysis of atmospheric waves
 - A. Scale analysis
 - B. perturbation method
 - C. Euler method
 - D. Leapfrog method